

01

Code specifications

Dataflow

Feature	DY	ttjets	QCD	Data
Sample Type	Drell-Yan MC	tt MC	Data-driven QCD	data files with the signal-like selection
Luminosity block filtering			✓	
Extra gen matching	✓	✓		
Generator weight	1.81	0.15		
HLT version	HLT_Ele32_WPTight_Gsf	HLT_Ele32_WPTight_Gsf	HLT_Ele32_WPTight_Gsf and Electron_mvaFall17_V2Iso_WP80	

Fase0: general description

- Tau ID variables (DeepTau),
- Tau momenta (pt, eta, phi, mass),
System variables (mass, pt, acoplanarity, rapidity),
- Jet info (pt, b-tagging),
- MET (missing transverse energy),
Generator-level info.

Calculates:

- Invariant mass of the tau pair,
- Acoplanarity between taus,
- System pt and rapidity,
- Number of b-tagged jets (DeepB > 0.4506),

Applies trigger filter (`HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg == 1`)

ETau Channel

Being used as control to understand the processing of the data samples;

No information available on what are the original samples used for this channel

Available samples:

- `Dados 2018 UL skimmed ETau nano fase0 merge.root`
- `Dados_2018_UL_skimmed_ETau_nano_fase1total_2018.root`
- `Dados_2018_UL_skimmed_ETau_nano_fase1total-protons_2018.root`

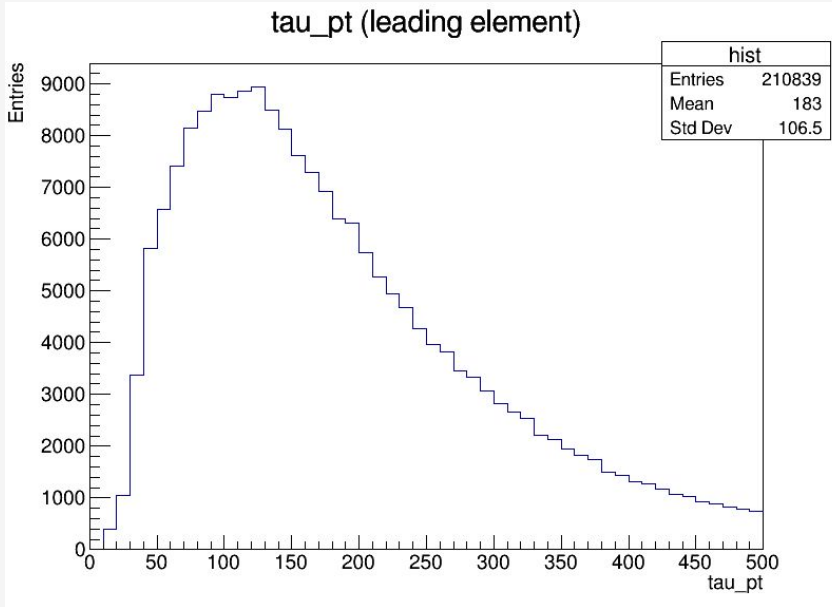
This channel can be used as “control” for the processing of these samples

ETau Channel -> Dados fase0 merge : available variables

proton_xi	- tau_id_antij	
- proton_thy	- tau_id_full	- e_pt
- proton_thx	- tau_decay	- e_eta
- proton_y	- tau_charge	- e_phi
- proton_x	- jet_pt	- e_energy
- proton_t	- jet_eta	- e_id
- proton_arm	- jet_phi	- e_charge
- proton_time	- jet_mass	- nVertices
- proton_trackx1	- jet_btag	- weight_sm
- proton_tracky1	- met_pt	- bsm_weights
- proton_trackx2	- met_phi	- mc_pu_trueinteractions
- proton_tracky2	- met_significance	
- proton_trackrpid1	- gen_proton_px	
- proton_trackrpid2	- gen_proton_py	
- proton_trackthx1	- gen_proton_pz	
- proton_trackthy1	- gen_proton_xi	
- proton_trackthx2	- gen_proton_t	
- proton_trackthy2	- gen_tau_pt	
- tau_energy	- gen_tau_eta	
- tau_pt	- gen_tau_phi	
- tau_eta	- gen_tau_energy	
- tau_phi	- gen_e_pt	
- tau_id	- gen_e_eta	
- tau_id_antimu	- gen_e_phi	
- tau_id_antie	- gen_e_energy	

ETau Channel -> Dados fase0 merge : available variables

In this file the branches are saved as vector <int> or vector<float>

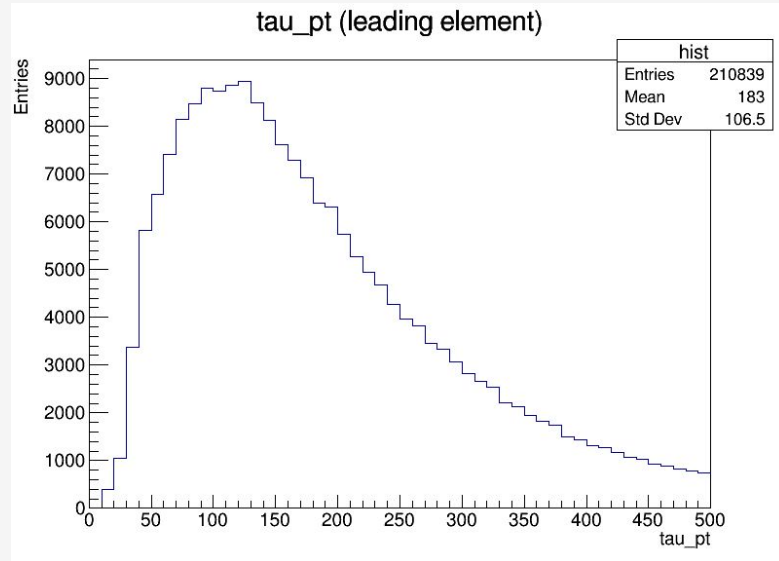


Plot to check that the branches weren't empty

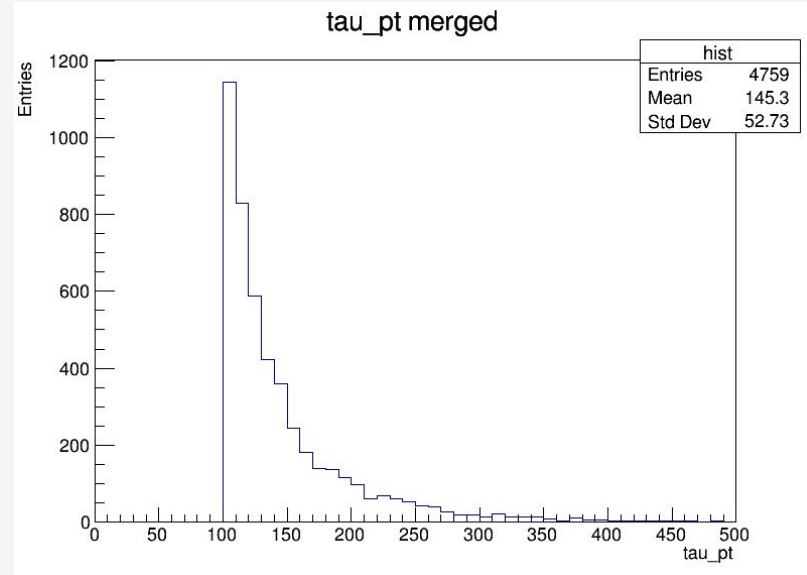
Since the branches are different and there is no reference, it is harder to create the phase1 analyser

ETau Channel: Data

Before phase 1 data analyser



After phase 1 data analyser



Current goal: create the code that produces these results.

Why?

Because in the MuTau channel the samples for phase1 AFTER the analyser are not available -> the goal is to create them to proceed with analysis

Code organization Background -> ETau Channel

Goal: only one file for signal and one file for background

(with parallelization using rootEnableImplicit)

File name	reads		Output
DY_fase0	DY 2018 UL.txt		DY 2018 UL skimmed Tau Tau nano
QCD_fase0	QCD 2018 UL.txt dadosluminosidade.txt	Checks if luminosity in event matches with the data from file	QCD 2018 UL skimmed Ta uTau nano
ttJets_fase0	ttJets 2018 UL.txt		ttJets 2018 UL skimmed TauTau nano
nanotry_fase1_DY	DY 2018.txt Name of fase0 output files		
nanotry_fase1_QCD	QCD fase1 PIC.txt Name of fase0 output files		
nanotry_fase1_ttJets	ttJets fase1 PIC.txt Name of fase0 output files		